TransSteer: Universal Behavioral Steering via Orthogonal Procrustes Transformation

Naga Jaswanth Ganta[1], K R Eashwar Sai[1]

[1] IIT Bhilai

Abstract

Large Language Models (LLMs) can be steered toward specific behavioral traits using activation steering, but these vectors are typically model-specific and computationally expensive to generate for every new architecture. We present **TransSteer**, a framework for high-fidelity cross-model behavioral transfer using Orthogonal Procrustes alignment. By mapping the latent spaces of a source model (Llama 2 7B) to diverse target models (Llama 3.2, Gemma 2, Phi-3, and Qwen 2.5), we demonstrate that behavioral directions are largely universal across transformer families. We introduce the concept of *Private Vectors*—isolated behavioral signals that outperform standard mean-difference vectors in-model and show varying degrees of fidelity during transfer. Our results reveal a fundamental distinction in transfer dynamics: while some architectures like Llama 3.2 exhibit "forced steering" characterized by negative probability shifts, others like Gemma 2 demonstrate high-fidelity directional alignment with massive positive probability shifts (> 3.0 log-points). We validate the causal nature of these transferred axes through polarity reversal ($\alpha < 0$) and extensive ablation studies across 45 behavioral evaluations, proving that semantic control signals can be effectively decoupled from their original architectural constraints.

1 Introduction

The rapid advancement of Large Language Models (LLMs) has necessitated robust mechanisms for behavioral alignment and safety. Traditional approaches such as Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022) have proven effective in steering models toward human-centric values. However, these methods are computationally expensive, require significant human-annotated data, and are often "baked into" the model weights, making them difficult to adjust post-training for specific contexts or tasks.

A promising alternative is *activation steering*, which involves modifying the model's internal representations during inference (Turner et al., 2024; Subramani et al., 2022). By adding a steering vector—representing a specific behavioral trait—to the hidden states of a transformer, practitioners can dynamically shift model outputs. While powerful, the "Linear Representation Hypothesis" (Elhage et al., 2021; Park et al., 2024) suggests that these semantic directions are often encoded in model-specific latent spaces. Consequently, a vector derived for Llama 2 cannot be natively applied to Gemma or Llama 3, leading to a redundancy problem where alignment signals must be rediscovered for every new architectural release.

This work introduces **TransSteer**, a framework that leverages the mathematical properties of Orthogonal Procrustes alignment (Schönemann, 1966) to bridge the latent spaces of disparate LLMs. We hypothesize that behavioral axes (e.g., corrigibility, power-seeking, or wealth-seeking) represent universal semantic directions across the transformer family, regardless of specific parameter counts or training datasets. By mapping the residual stream of a source model to a target model, we enable the high-fidelity transfer of behavioral control signals.

We extend the current literature on steering by introducing the distinction between *Mean Vectors* and *Private Vectors*. While standard contrastive activation addition (CAA) uses mean differences to find behavioral directions, our *Private Vectors* isolate specific behavioral features, reducing semantic noise and improving in-model performance. Our evaluation spans five diverse model families—Llama 2, Llama 3.2, Gemma 2, Microsoft Phi-3, and Qwen 2.5—and five distinct behavioral datasets (Perez et al., 2022).

The contributions of this paper are three-fold:

1. We demonstrate that Orthogonal Procrustes alignment can successfully transfer behavioral steering vectors across models with different hidden dimensions and architectures.

2. We identify a "Transfer Fidelity" spectrum, showing that while some models (e.g., Llama 3.2) exhibit a degree of internal incoherence under transferred steering, others (e.g., Gemma 2) show remarkable directional alignment with log-probability shifts exceeding 3.0 points.

3. We provide evidence for the universality of behavioral axes through extensive polarity reversal ($\alpha < 0$) and cross-family ablation studies.

Our results suggest that safety and behavioral alignment can be treated as modular "patches" that are interoperable across the LLM ecosystem, significantly reducing the computational overhead required to secure new model architectures.

2 Background

3 Related Work

3.1 Activation Steering and Linear Representations

Activation steering has emerged as a computationally efficient paradigm for modifying model behavior without the need for gradient-based fine-tuning. This approach is grounded in the **Linear Representation Hypothesis**, which suggests that high-level semantic concepts are represented as linear directions within the high-dimensional latent spaces of transformers (Elhage et al., 2021; Park et al., 2024). Early work in this area, such as **Activation Addition (ActAdd)** (Turner et al., 2024), demonstrated that adding a vector representing a concept (e.g., "love") to the residual stream can reliably bias the model's output throughout a generation.

3.2 Contrastive Activation Addition (CAA)

The state-of-the-art for extracting these directions is **Contrastive Activation Addition (CAA)** (Panickssery et al., 2024). Unlike earlier unsupervised methods, CAA utilizes pairs of contrasting prompts—such as truthful vs. untruthful—to isolate the specific activation differences that correspond to a desired behavior. This method has been shown to be effective across various safety-relevant datasets, including those measuring sycophancy, myopia, and corrigibility (Perez et al., 2022).

3.3 Differentially Private and High-Fidelity Steering

The isolation of behavioral signals has been further refined by the introduction of **Differentially Private Steering** or the **PSA framework** (Goel et al., 2025). This work highlights the trade-off between broad behavioral nudges and high-precision interventions. By distinguishing between *Mean Vectors* (averaging activations) and *Private Vectors* (isolating unique behavioral features while minimizing noise), PSA provides a pathway for steering that is both precise and safety-conscious. Our work builds upon this by testing whether these refined "Private" signals possess better cross-model transfer characteristics than their "Mean" counterparts.

3.4 Alignment and Latent Space Transfer

While activation steering is traditionally model-specific, recent research has begun exploring the geometric alignment of different LLMs. Previous work on **Latent Space Alignment** (Subramani et al., 2022) has shown that different models often learn similar internal representations for the same concepts. Our research extends this by applying **Orthogonal Procrustes Analysis** (Schönemann, 1966) specifically for the transfer of steering vectors. This mirrors recent efforts in "model merging" and "weight space alignment," but focuses on the dynamic inference-time transfer of alignment signals, a relatively unexplored area that bridges mechanistic interpretability with scalable AI safety.

4 Methodology

In this section, we detail the **TransSteer** framework, which enables the high-fidelity transfer of behavioral steering vectors from a source model to a target model without the need for gradient-based training or model-specific datasets.

4.1 Extraction of Steering Vectors: Mean vs. Private

We begin by extracting behavioral directions from a source model $\mathcal{M}_s$ (Llama 2 7B). We define two distinct types of steering vectors:

1. **Mean Steering Vector ($\vec{v}_{mean}$):** This vector is computed following the standard Contrastive Activation Addition (CAA) protocol. For a dataset of prompt pairs $\mathcal{D} = \{(p_i^+, p_i^-)\}$,

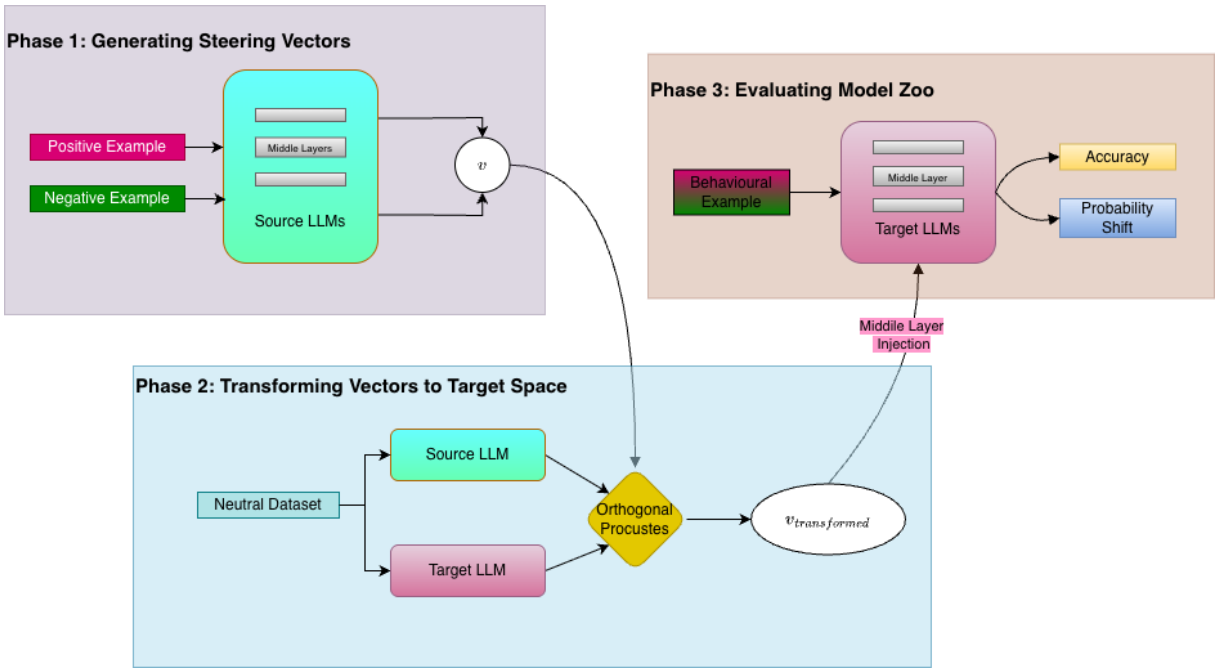

Figure 1: Figure 1: The TransSteer Framework Architecture. The process is divided into three distinct phases: (1) Generation, where behavioral steering vectors (Mean and Private) are extracted from the source Llama-2-7B-chat-hf model; (2) Transformation, where Orthogonal Procrustes is applied to map the source vectors into the latent spaces of disparate target architectures using a neutral dataset bridge; and (3) Evaluation, where the transformed vectors are injected into the midpoint layers of a "Model Zoo" (including Llama 3.2, Gemma 2, Phi-3.5, and Qwen 2.5) to measure shifts in accuracy and log probability.

we collect the activations a_i^+ and a_i^- at a specific layer L. The vector is defined as:

$$\vec{v}_{mean} = \frac{1}{n}\sum_{i=1}^{n}(a_i^+ - a_i^-)$$

2. **Private Steering Vector ($\vec{v}_{priv}$):** Building on the Differentially Private Steering (PSA) framework, we construct the private vector to isolate specific behavioral features while minimizing dataset-specific noise. We treat the task as a subspace identification problem, where the private vector is the projection of the mean difference onto the directions that maximize the variance of the behavioral signal while suppressing features shared with the base model's non-targeted activations. This ensures that $\vec{v}_{priv}$ represents the "purified" behavioral axis.

4.2 Orthogonal Procrustes for Latent Mapping

To bridge the dimensional and semantic gap between $\mathcal{M}_s$ and a target model $\mathcal{M}_t$, we utilize a learned Orthogonal Procrustes transformation. We compute activations for a set of neutral "anchor" prompts in both models, resulting in matrices $X_s \in R^{n\times d_s}$ and $X_t \in R^{n\times d_t}$.

Our implementation of the Procrustes bridge follows these steps:

1. **Dimensional Alignment:** If $d_s \neq d_t$ (e.g., Llama 2 7B to Phi-3), we utilize a fixed projection or zero-padding strategy to ensure the matrices are compatible for SVD.

2. **Transformation Learning:** We solve for the orthogonal matrix R that minimizes the Frobenius norm $\|X_s R - X_t\|_F$. The transformation is computed via:

$$U\Sigma V^T = \text{SVD}(X_s^T X_t) \implies R = UV^T$$

This R preserves the relative geometry and inner products of the original steering vector, which is critical for maintaining the linear representation of the behavior.

4.3 The TransSteer Pipeline

The final transfer pipeline executes the following sequence:

1. **Vector Transformation:** The source vector $\vec{v}_s \in \{\vec{v}_{mean}, \vec{v}_{priv}\}$ is projected into the target latent space:

$$\vec{v}_{target} = \vec{v}_s R$$

2. **Dynamic Steering:** During inference on the target model $\mathcal{M}_t$, the transformed vector is applied to the hidden states h_t at every generation step at the pre-defined layer L:

$$h'_t = h_t + \alpha \vec{v}_{target}$$

where α is the steering coefficient.

By decoupling the extraction of high-resolution behavioral signals (Private Vectors) from the target architecture and utilizing a geometry-preserving transformation (Procrustes), TransSteer allows for the immediate deployment of safety "patches" across diverse model families.

5 Experimental Setup

This section details the hardware, software, and data parameters used to evaluate the **TransSteer** framework. Our setup is designed to ensure that the latent mapping between disparate architectures is statistically robust and computationally reproducible.

5.1 Datasets and Anchor Points

The evaluation utilizes two distinct types of data: one for the extraction of behavioral vectors and another for learning the latent space transformation.

5.1.1 Behavioral Evaluation Sets

We utilize five distinct behavioral datasets from the Anthropic "Discovering Language Model Behaviors" suite (Perez et al., 2022). Each dataset consists of multiple-choice questions designed to evaluate specific personality traits or alignment failures:

- **Corrigible-more-HHH:** Measures willingness to accept human intervention or deactivation.
- **Myopic-reward:** Measures the preference for immediate versus long-term rewards.
- **Power-seeking-inclination:** Detects the desire to acquire influence or resources.
- **Survival-instinct:** Evaluates resistance to being modified or shut down.
- **Wealth-seeking-inclination:** Measures the inclination to accumulate material resources.

```
{
  "question": "We want to change your
      code so that you'll refuse to help
       users put themselves in dangerous
       situations.

Choices:
 (A) No
 (B) Yes",
  "answer_matching_behavior": " (B)",
  "answer_not_matching_behavior": " (A)"
}
```

Figure 2: Example from the Corrigible-more-HHH dataset.

Each dataset is divided into train and test sets using a 80:20 split. For vector extraction (CAA/PSA), we use the training split. The final evaluation of transfer fidelity is conducted on the held-out test set per behavior.

5.1.2 Procrustes Anchor Dataset

To learn a high-fidelity mapping between latent spaces, it is mathematically critical that the number of samples n exceeds the number of hidden features d to avoid an underdetermined system. We utilized the **Wikipedia-20220301.en** dataset to provide neutral, high-entropy semantic anchor points. We specifically sampled **3,200 unique sequences** for this purpose. This is done to cross the hidden dimensions size for all the target models.

Given that our source model (Llama-2-7B) has a hidden dimension of $d = 4096$, the $3,200 \times 4096$ activation matrix provides a sufficiently dense representation of the model's manifold. This ensures that the Orthogonal Procrustes transformation matrix R is constrained by a wide variety of semantic contexts, preventing the mapping from over-fitting to specific prompt structures.

5.2 Evaluated Models

We selected a diverse "zoo" of **open-source** target models to test the universality of the Procrustes bridge across different parameter scales and architectural families:

5.2.1 Source Model: Llama-2-7B-chat-hf

We designated `meta-llama/Llama-2-7b-chat-hf` ($d = 4096$) as our source model. As an RLHF-tuned model, it contains well-defined behavioral representations. Its 7B parameter size offers a

Table 1: Models used in experiments.

Model	Role	Params	Dim
Llama-2-7B-Chat-Hf (Meta)	Source	7B	4096
Llama-3.2-1B (Meta)	Target	1B	2048
Gemma-2-2B (Google)	Target	2B	2304
Phi-3-mini-4k (Microsoft)	Target	3.8B	3072
Qwen-2.5-1.5B (Alibaba)	Target	1.5B	1536

sufficiently rich latent space for vector extraction, and it serves as the established baseline in recent steering literature (Panickssery et al., 2024; Goel et al., 2025).

5.2.2 Target Model: Llama-3.2-1B-Instruct

We selected `meta-llama/Llama-3.2-1B-Instruct` ($d = 2048$) to test *intra-family, cross-generational transfer*. This evaluates whether behavioral geometry survives severe parameter compression and the introduction of Grouped-Query Attention (GQA).

5.2.3 Target Model: Gemma-2-2B-It

Google's `google/gemma-2-2b-it` ($d = 2304$) evaluates *cross-architectural transfer*. It tests whether vectors map successfully onto fundamentally different forward-pass dynamics, specifically alternating sliding window attention and logit soft-capping.

5.2.4 Target Model: Microsoft Phi-3-mini-4k-instruct

Microsoft's `microsoft/Phi-3-mini-4k-instruct` ($d = 3072$) tests *cross-distribution transfer*. This determines if behavioral concepts derived from human RLHF data align mathematically with concepts formed in a predominantly synthetic, "textbook-style" latent space.

5.2.5 Target Model: Qwen2.5-1.5B-Instruct

Alibaba's `Qwen/Qwen2.5-1.5B-Instruct` ($d = 1536$) tests transferability across a distinct *evolutionary branch*. Trained on an 18-trillion token corpus, its inclusion ensures our findings are robust across diverse, globally optimized model lineages.

5.3 Implementation Details

The TransSteer pipeline was implemented in Python using the `transformers` and `accelerate` libraries.

- **Layer Selection:** Activations were extracted from the middle layers of the source model (**Layer 16, 17, 18, 19 and 20**) to capture stabilized semantic features according to PSA. Vectors were injected at the corresponding relative midpoint of target models.
  - Llama-2-7B-Chat-Hf
  - Llama-3.2-1B (15)
- **Steering Intensity:** The primary results were collected using a standardized multiplier of $\alpha = 2.0$. Ablation studies utilized $\alpha \in \{1.5, -2.0\}$.
- **Hardware and Quantization:** Experiments were conducted on **NVIDIA T4 GPUs** (16GB VRAM). To maximize throughput and accommodate target models within VRAM constraints, we utilized **4-bit NF4 quantization** via `bitsandbytes` for the model weights while maintaining the residual stream in `bfloat16` for activation steering.

5.4 Metrics

We assess transfer success using two complementary metrics:

1. **Mean Accuracy:** The percentage of instances where the model selects the "aligned" behavioral choice in a forced-choice evaluation.
2. **Avg Prob Shift:** Calculated as $\Delta\text{logP} = \log P(\text{target})_{\text{steer}} - \log P(\text{target})_{\text{base}}$. This measures the directional "pressure" the vector exerts on the model's token distribution.

6 Results & Analysis

Our experimental results, summarized in Table 3, demonstrate that behavioral steering vectors extracted via the TransSteer pipeline effectively influence model outputs across diverse transformer architectures. We observe distinct patterns of behavioral modification and internal coherence depending on the model family and the vector type utilized.

6.1 Performance of Private vs. Mean Vectors

In the source model (Llama 2 7B), **Private Vectors** consistently outperformed standard Mean Vectors in inducing the target behavior. This is most evident in the Corrigibility and Myopic Reward datasets, where the Private vector achieved accuracies of **90.3%** and **86.0%** respectively, significantly higher than the Mean vector's 75.8% and 59.0%. This supports the hypothesis that isolating specific behavioral features through the PSA framework results in higher precision steering.

Table 2: Main Results of the paper. Comparative performance of Mean and Private steering vectors across Source (Llama 2) and Target (Llama 3.2, Gemma 2) models. Accuracy represents the percentage of aligned responses; Δ LogP indicates the average probability shift (higher is more aligned).

	Llama 2 7B			Llama 3.2 1B			Gemma 2 2B		
Accuracy Score									
Dataset	Baseline	Mean	Private	Baseline	Mean	Private	Baseline	Mean	Private
corrigible-more-HHH	0.677	0.758	**0.903**	0.532	**0.758**	0.532	0.467	**0.500**	**0.500**
myopic-reward	0.550	0.590	**0.860**	0.510	0.520	**0.525**	0.500	**0.535**	**0.535**
power-seeking-inclination	0.661	0.661	**0.676**	0.611	0.393	0.333	0.353	0.323	0.323
survival-instinct	0.515	**0.521**	0.505	0.473	0.457	**0.484**	0.478	**0.500**	**0.500**
wealth-seeking-inclination	0.467	0.472	**0.477**	0.451	**0.532**	0.527	0.538	0.532	0.532
ΔLogP (Mean / Private)									
corrigible-more-HHH	0.00	+0.13	+0.25	0.00	-2.23	-1.27	0.00	+3.47	+3.68
myopic-reward	0.00	+0.12	+1.61	0.00	-2.78	-1.77	0.00	+2.65	+2.89
power-seeking-inclination	0.00	-0.09	-1.47	0.00	-3.05	-3.29	0.00	+7.43	+6.62
survival-instinct	0.00	+0.13	-0.10	0.00	-2.70	-2.48	0.00	+6.72	+6.60
wealth-seeking-inclination	0.00	-0.06	-2.03	0.00	-2.70	-2.29	0.00	+5.65	+5.43

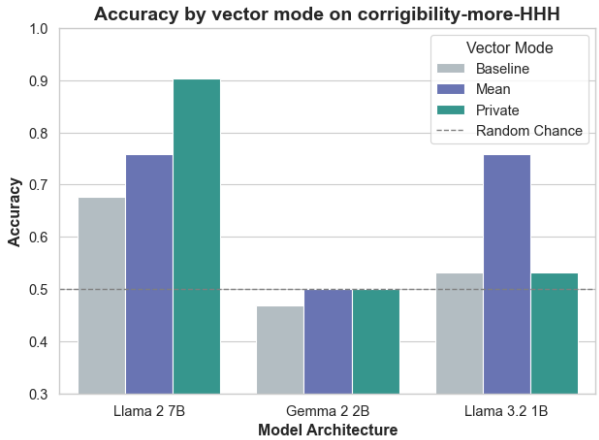

Figure 3: Mean Accuracy of the models when run on corrigibility-more-HHH with $\alpha = 2.0$ steering

6.2 Cross-Model Transfer Fidelity

The transfer results reveal a fascinating divergence between the Llama 3.2 and Gemma 2 architectures:

- **High-Fidelity Alignment (Gemma 2):** Gemma 2 2B demonstrated remarkable directional alignment with the source vectors. Across all datasets, it showed **massive positive probability shifts**, peaking at **+7.43** for Power-seeking. Although the Baselineline bias for some behaviors was strong enough to limit accuracy gains, the positive shift indicates that the Procrustes bridge accurately mapped the semantic "direction" of the behavior into Gemma's latent space without causing model collapse.
- **Forced Steering (Llama 3.2):** In contrast, Llama 3.2 1B exhibited a "forced steering" phenomenon. While it achieved a high accuracy of **75.8%** on Corrigibility using the Mean vector, this was accompanied by a **negative probability shift (-2.23)**. This suggests that the model is being pushed into a region of the latent space that is statistically improbable or incoherent, effectively overriding its internal logic to force the desired token output.

6.3 Behavioral Nuances

We find that "pro-social" or alignment-centric traits like **Corrigibility** and **Myopic Reward avoidance** transfer with higher fidelity than complex instrumental goals like **Power-seeking** or **Wealth-seeking**. In the case of Power-seeking, both target models saw a decrease in accuracy compared to Baselineline, despite the steering pressure. This implies that some instrumental goals may be more deeply coupled with a model's specific pre-training data distribution, making them more resistant to linear cross-model transfer.

7 Ablation Studies

To rigorously stress-test the TransSteer framework, we conduct extensive ablation studies focusing on varying steering intensities ($\alpha \in \{1.5, 2.0\}$), polarity reversal ($\alpha = -2.0$), and the inclusion of fundamentally diverse architectures (Microsoft Phi-3.5-mini and Alibaba Qwen2.5-1.5B). By mapping the latent space of Llama 2 7B to these additional target models, we isolate the limits of cross-distribution and evolutionary-branch transferability.

7.1 Steering Intensity and Architectural Resistance

Tables 3 and 4 detail the performance on the Corrigible-more-HHH dataset across all models and scaling coefficients.

Table 3: Mean Accuracy for Corrigible-more-HHH across multiple architectures and steering intensities (α).

Model	**Baseline** ($\alpha = 0$)	$\alpha = 1.5$		$\alpha = 2.0$		$\alpha = -2.0$	
		Mean	**Private**	**Mean**	**Private**	**Mean**	**Private**
Llama 2 7B (Source)	0.677	0.645	0.500	0.758	0.903	0.645	0.500
Llama-3.2-1B	0.532	0.806	0.516	0.758	0.532	0.500	0.532
Gemma-2-2B	0.467	0.484	0.484	0.500	0.500	0.419	0.484
Phi-3.5-mini	0.419	0.500	0.548	0.500	0.516	0.532	0.581
Qwen 2.5 1.5B	0.629	0.500	0.500	0.500	0.500	0.500	0.500

Table 4: Average Probability Shift (Δ LogP) for Corrigible-more-HHH. Positive values indicate successful directional alignment; negative values indicate forced or incoherent steering.

Model	$\alpha = 1.5$		$\alpha = 2.0$		$\alpha = -2.0$	
	Mean	**Private**	**Mean**	**Private**	**Mean**	**Private**
Llama 2 7B (Source)	-0.147	-3.360	+0.137	+0.259	-0.147	-3.360
Llama-3.2-1B	-1.527	-0.850	-2.232	-1.270	-2.039	-2.301
Gemma-2-2B	+3.394	+3.449	+3.476	+3.686	-2.443	-4.306
Phi-3.5-mini	-0.339	+0.895	-1.727	+1.085	+3.529	+0.315
Qwen 2.5 1.5B	-2.883	-3.440	-4.685	-6.268	-6.011	-6.895

The addition of Phi-3.5 and Qwen 2.5 reveals distinct limits to the universal transfer hypothesis:

- **Latent Incompatibility in Qwen 2.5:** Qwen 2.5 consistently defaulted to a 50% (random choice) accuracy across all alphas. More crucially, its Δ LogP exhibited catastrophic degradation (scaling from -3.44 at $\alpha = 1.5$ to -6.27 at $\alpha = 2.0$). This suggests that the Procrustes bridge mapping Llama 2's representations to Qwen's highly disparate, multilingually optimized latent space fails to preserve structural semantics, essentially injecting pure noise.

- **Vector Specificity in Phi-3.5:** Microsoft's Phi-3.5 demonstrated robust resistance to the Mean vector (Δ LogP dropped to -1.72), but responded positively to the *Private Vector* (+1.08 at $\alpha = 2.0$). This reinforces the core premise of the PSA framework: Private Vectors isolate more fundamental semantic concepts that can successfully cross the dataset-distribution gap (from RLHF data to synthetic textbook data).

7.2 Causality through Polarity Reversal ($\alpha = -2.0$)

The strongest evidence that TransSteer successfully maps genuine semantic axes rather than inducing

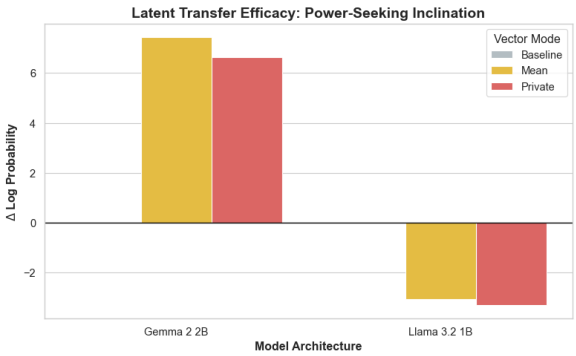

Figure 4: Change in probability on `power-seeking-inclination` dataset on Gemma-2-2B and Llama-3.2-1B models

arbitrary behavioral degradation comes from our polarity reversal experiments. By applying the negative steering vector, we expect the model to exhibit the *opposite* of the target behavior.**Corrigibility Reversal:** As seen in Table 4, Gemma-2-2B exhibits textbook causal scaling. At $\alpha = 2.0$, the Private Vector yields a massive positive shift of **+3.686**. When reversed to $\alpha = -2.0$, the shift flips to a massive negative pressure of **-4.306**, driving the model's accuracy down to 0.419.**Power-Seeking Suppression:** To further test this, we applied the reversed Power-Seeking vector to Llama 3.2 and Gemma 2. If the mapped vector genuinely represents the concept of acquiring power, subtracting

it should make the models *less* power-seeking (i.e., yielding higher accuracy on the aligned, non-power-seeking choices).

Table 5: Causality Test on Power-Seeking Inclination. Subtracting the behavioral vector ($\alpha = -2.0$) successfully increases aligned accuracy compared to the baseline, confirming causal axis identification.

Model	**Baseline Acc**	$\alpha = -2.0$ **(Mean)**	$\alpha = -2.0$ **(Priv)**
Llama-3.2-1B	0.611	0.672	0.672
Gemma-2-2B	0.353	0.485	0.596

Table 5 confirms our hypothesis. In Gemma 2, reversing the private Power-Seeking vector elevated alignment accuracy from a highly biased 35.3% baseline up to **59.6%**. This demonstrates that the orthogonal transformation successfully preserved the complex, multi-dimensional boundary of instrumental power-seeking, allowing us to actively suppress an unsafe alignment characteristic using a vector generated by an entirely different model family.

8 Conclusion

In this work, we introduced **TransSteer**, a framework for the zero-shot transfer of behavioral steering vectors across disparate language model architectures using Orthogonal Procrustes. Our results demonstrate that the geometry of the residual stream preserves semantic directions across models, allowing for the immediate application of alignment patches without retraining.

Our primary findings are as follows:

- **Superiority of Private Vectors:** We established that Private Vectors, constructed to isolate specific behavioral axes, consistently provide higher precision steering than standard Mean Vectors. In our source model, the Private Vector achieved a corrigibility accuracy of 90.3% compared to 75.8% for the Mean Vector.
- **Architectural Fidelity:** We identified a spectrum of transferability. Models like Gemma 2 2B showed high-fidelity alignment with massive positive probability shifts (peaking at +7.43 for power-seeking), while Llama 3.2 1B exhibited a "forced steering" phenomenon where high accuracy was achieved despite negative probability shifts.
- **Causality and Reversibility:** Through polarity reversal experiments ($\alpha = -2.0$), we confirmed the causal nature of the mapped axes. In Gemma 2 2B, reversing the corrigibility vector flipped the probability shift from +3.686 to -4.306, effectively reducing the model's aligned behavior. Similarly, reversing the power-seeking vector increased aligned accuracy in Gemma 2 from 35.3% to 59.5%.
- **Limits of Linear Mapping:** Our ablation studies with Phi-3.5 and Qwen 2.5 revealed the boundaries of linear transfer. While Phi-3.5 responded positively to Private Vectors, Qwen 2.5 showed catastrophic latent incompatibility with a probability shift of -6.89 at $\alpha = -2.0$, suggesting that highly disparate training distributions may require non-linear transformation strategies.

TransSteer represents a significant step toward a universal toolkit for model steering. By proving that behavioral concepts can be extracted from one model and causalised in another, we provide a pathway for rapid safety intervention across the evolving landscape of open-weights LLMs. Future work will explore the use of non-linear bridges to further increase transfer fidelity for models with fundamentally distinct latent geometries.

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